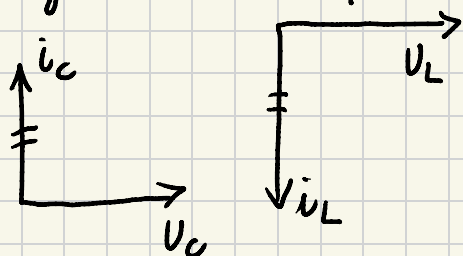


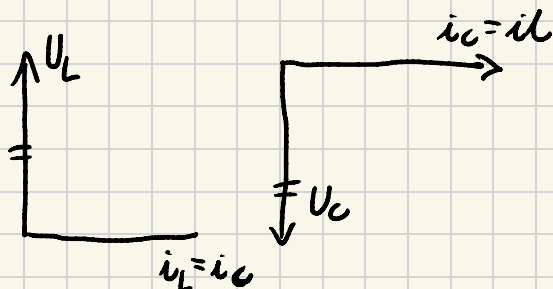
WYKŁAD 6

Rezonans w obwodach RLC

Rezonans charakteryzuje się tym, że pomimo wystąpienia L i C , obwód zachowuje charakter rezystancyjny. L i C wzajemnie się znoszą.



Równoległe

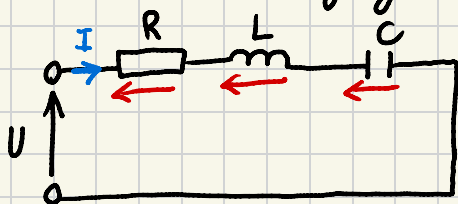


Szeregowy

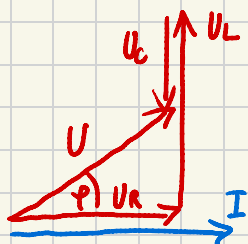
Rezonans

- 1) Szeregowy (połączenie szeregowe L i C)
- 2) Równoległy (połączenie równoległe L i C)

1) Rezonans szeregowy



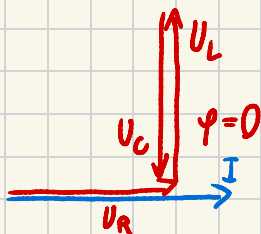
$$u(t) = U_m \sin(\omega t + \varphi)$$



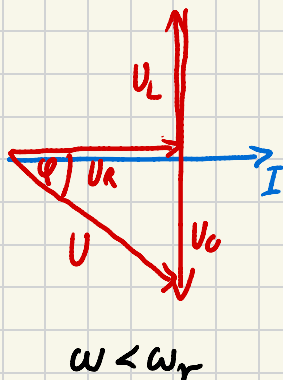
$$U_L = j\omega L \cdot I$$

$$U_C = -j\frac{1}{\omega C} \cdot I$$

$$\omega > \omega_r$$



charakter
rezystancyjny
(punkt rezonansu)
 $\omega = \omega_r$



$$\omega < \omega_r$$

Warunek rezonansu szeregowego: $\text{Im}(Z_{RLC}) = 0$

$$\omega L \cdot I = \frac{1}{\omega C} I \quad \wedge \quad \omega = \omega_r$$

$$\omega_r^2 = \frac{1}{LC}$$

$$\omega_r = \frac{1}{\sqrt{LC}}$$

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$U = R \cdot I_R \quad I_R = \frac{U}{R}$$

$$|U_{Lr}| = \omega_r L \cdot \frac{U}{R} = U \cdot \frac{\omega_r L}{R}$$

$$|U_{Cr}| = \frac{1}{\omega_r C} \cdot \frac{U}{R} = U \cdot \frac{1}{\omega_r C R}$$

Dobroć obwodu rezonansowego w punkcie rezonansu

$$Q = \frac{\omega_r \cdot L}{R} = \frac{\frac{1}{\sqrt{LC}} \cdot L}{R} = \frac{\sqrt{\frac{L}{C}}}{R}$$

$$\omega_r = \frac{1}{\sqrt{LC}}$$

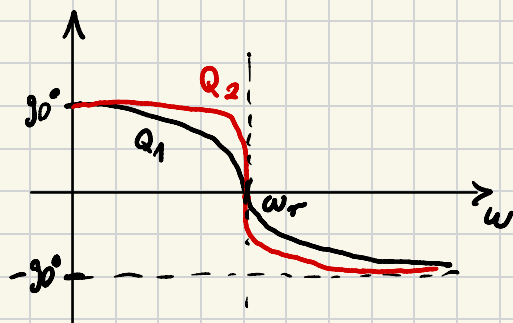
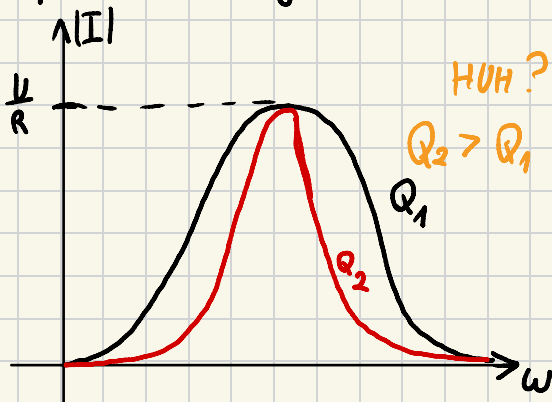
$$Q = \frac{1}{\omega_r C R} = \frac{\sqrt{LC}}{C R} = \frac{\sqrt{\frac{L}{C}}}{R}$$

Charakterystyki częstotliwościowe obwodu rezonansowego

$$I = \frac{U}{R + j\omega L - j\frac{1}{\omega C}} = \frac{U}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2} \cdot e^{j \arctg(\frac{\omega L - \frac{1}{\omega C}}{R})}}$$

$$|I(\omega)| = \frac{|U|}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}} \rightarrow \text{amplitudowa charakterystyka}$$

$$\varphi(\omega) = -\arctg \frac{(\omega L - \frac{1}{\omega C})}{R} \rightarrow \text{fazowa charakterystyka}$$



$$U_L = j\omega L \cdot I = \omega L \cdot I \cdot e^{j90^\circ}$$

$$|U_L| = \frac{\omega L \cdot U}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}}$$

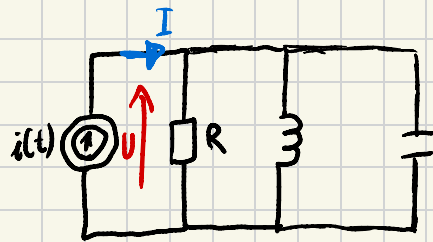
$$\varphi_{U_L} = 90^\circ + \arctg \frac{(\omega L - \frac{1}{\omega C})}{R}$$

$$U_C = -j \frac{1}{\omega C} I = \frac{1}{\omega C} \cdot I \cdot e^{-j90^\circ}$$

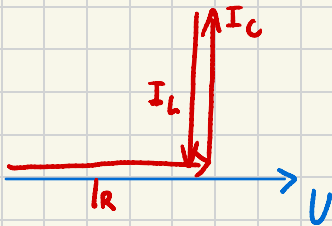
$$|U_C| = \frac{U}{\omega C \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}}$$

$$\varphi_{U_C} = -90^\circ - \arctg \left(\frac{\omega L - \frac{1}{\omega C}}{R} \right)$$

Rezonans r  ndleg  



$$i(t) = I_m \sin(\omega t + \varphi) \quad \varphi = 0$$

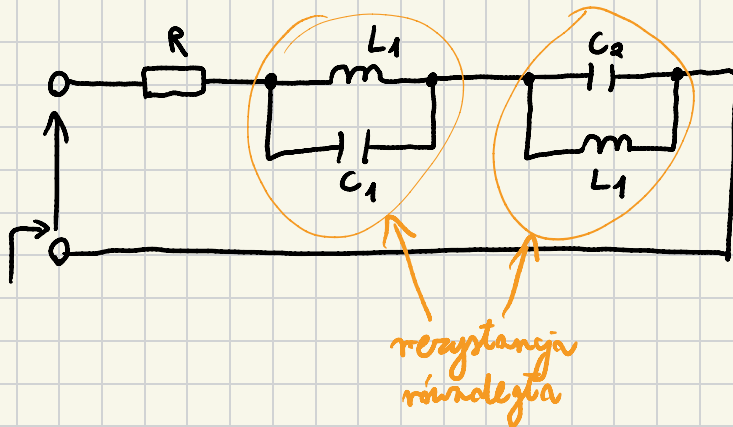


Warunki rezonansu

$$Z_n(Y) = (G + j\omega C - j\frac{1}{\omega L}) = 0$$

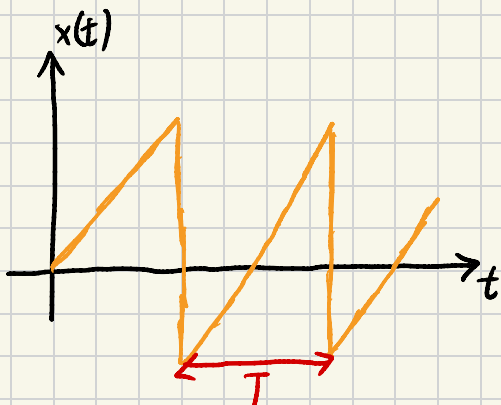
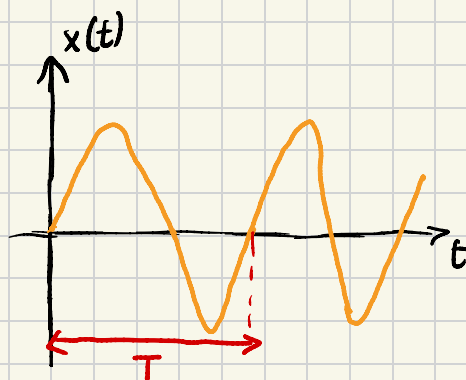
$$Z_n(Y) = 0$$

$$\omega_r C = \frac{1}{\omega_r L} \rightarrow \omega_r = \frac{1}{\sqrt{LC}}$$



$$\operatorname{Im}(Z) = 0 \quad \leftarrow \text{rezystancja szeregowo}$$

$$\operatorname{Im}(Y) = 0 \quad \leftarrow \text{rezystancja równoległa}$$



Sygnały $x(t)$ okresowe,

ale nie sinusoidalne!

SZEREG FOURIERA

Rozkład sygnałów niesinusoidalnych (okresowych) na sumę sygnałów sinusoidalnych o różnych częstotliwościach (harmonicznych)

$x(t)$ - sygnał okresowy niesinusoidalny

$$x(t) = A_0 + \sum_{k=1}^{\infty} [A_k \cos k\omega t + B_k \sin k\omega t] \quad \omega_k = k\omega$$

↑
k-ta harmoniczna

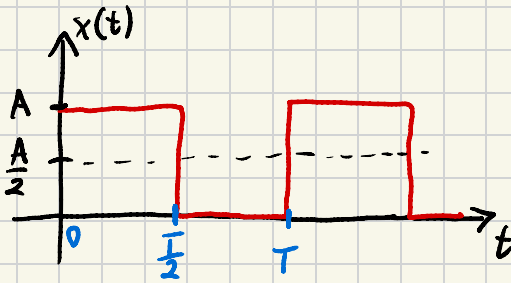
$$A_0 = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) dt$$

← DC (składowa stała)

$$A_k = \frac{2}{T} \int_{t_0}^{t_0+T} x(t) \cdot \cos k\omega t dt$$

$$B_k = \frac{2}{T} \int_{t_0}^{t_0+T} x(t) \cdot \sin k\omega t dt$$

Przykład:



$$A_0 = \frac{1}{T} \int_0^{\frac{T}{2}} A dt = \frac{A}{T} \cdot t \Big|_0^{\frac{T}{2}} = \frac{A}{T} \cdot \frac{T}{2} = \frac{A}{2}$$

$$A_k = \frac{2}{T} \int_0^{\frac{T}{2}} A \cos k\omega t dt = \frac{2A}{T} \cdot \frac{1}{k\omega} \cdot \sin k\omega t \Big|_0^{\frac{T}{2}} =$$

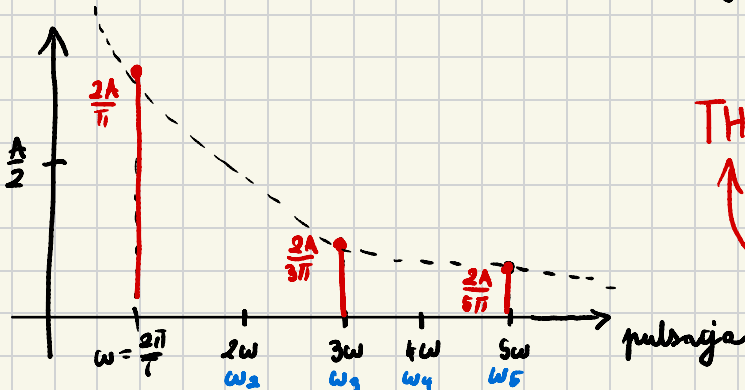
„sin(kπ) = 0“

$$= \frac{2A}{T k \omega} \left[\sin\left(\frac{k \cdot 2\pi}{T} \cdot \frac{T}{2}\right) - \sin(0) \right] = 0$$

$$B_k = \frac{2}{T} \int_0^{\frac{T}{2}} A \sin k\omega t dt = \frac{2A}{T} \cdot \frac{1}{k\omega} \cdot (-\cos k\omega t) \Big|_0^{\frac{T}{2}} =$$

$$= \frac{2A \cdot T}{T \cdot k \cdot 2\pi} [\cos 0 - \cos(k \cdot \frac{2\pi}{T} \cdot \frac{T}{2})] = \frac{A}{k\pi} [1 - \cos(k\pi)] = \begin{cases} \frac{2A}{k\pi} & k \text{ niep.} \\ 0 & k \text{ parzyste} \end{cases}$$

$$x(t) = \frac{A}{2} + \sum_{k=1}^{\infty} \frac{A}{k\pi} (1 - \cos k\pi) \cdot \sin \frac{k\omega t}{\omega_k} = \begin{cases} \frac{2A}{k\pi} & k. \text{ nieparzyste} \\ 0 & k \text{ parzyste} \end{cases}$$



$$THD = \frac{\sqrt{\sum_{k=1}^{\infty} |U_k|^2}}{|U_1|}$$

total harmonic difference

$$x(t) = A_0 + \sum_{k=1}^{\infty} [A_k \cos k\omega t + B_k \sin k\omega t]$$

$$x(t) = A_0 + \sum_{k=1}^{\infty} X_{mk} \cdot \sin(k\omega t + \psi_k)$$

$$X_{mk} = \sqrt{A_k^2 + B_k^2}$$

$$\psi_k = \arctg \frac{A_k}{B_k}$$

$$x(t) = A_0 + \sum_{k=1}^{\infty} X_{mk} \cdot \cos(k\omega t + \phi_k)$$

Niech:

$$x(t) = A_0 + \sum_{k=1}^{\infty} X_{mk} \cdot \cos(k\omega t + \phi_k)$$

$$x(t) = A_0 + \sum_{k=1}^{\infty} \frac{e^{jk\omega t} \cdot e^{j\phi_k} + e^{-jk\omega t} \cdot e^{-j\phi_k}}{2}$$

$$x(t) = A_0 + \sum_{k=1}^{\infty} \left(\frac{X_{mk}}{2} \cdot e^{j\phi_k} \right) \cdot e^{jk\omega t} + \left(\frac{X_{mk}}{2} e^{-j\phi_k} \right) \cdot e^{-jk\omega t}$$

$$x(t) = A_0 + \sum_{k=1}^{\infty} (X_k \cdot e^{jk\omega t} + X_k \cdot e^{-jk\omega t}) = A_0 + \sum_{\substack{k=-\infty \\ k \neq 0}}^{\infty} X_k \cdot e^{jk\omega t}$$

$$X_k = \frac{X_{mk}}{2} \cdot e^{j\phi_k}$$

$$|X_k| = \frac{X_{mk}}{2} \quad \varphi_k = \phi_k$$

Zupełna postać szeregu
Fouriera

Prägtad:

$$x(t) = 10 + 30 \sin \omega t + 24 \cos(2\omega t + 45^\circ)$$

$$x(t) = 10 + 30 \frac{e^{j\omega t} - e^{-j\omega t}}{2j} + 24 \frac{e^{j45^\circ} e^{j2\omega t} + e^{-j45^\circ} e^{-j2\omega t}}{2}$$

$$x(t) = 10 + 15 e^{-j90^\circ} e^{j\omega t} - 15 e^{-j90^\circ} e^{-j\omega t} + 12 e^{j45^\circ} e^{j2\omega t} + 12 e^{-j45^\circ} e^{-j2\omega t}$$

$$x(t) = 10 + 15 e^{-j90^\circ} e^{j\omega t} + 15 e^{j90^\circ} e^{-j\omega t} + 12 e^{j45^\circ} e^{j2\omega t} + 12 e^{-j45^\circ} e^{-j2\omega t}$$

